Access Control - Practical Lab

1. DAC - Discretionary Access Control

We created report.txt and used **chmod 740**. The owner has rwx, the group has read access, and others have no access.

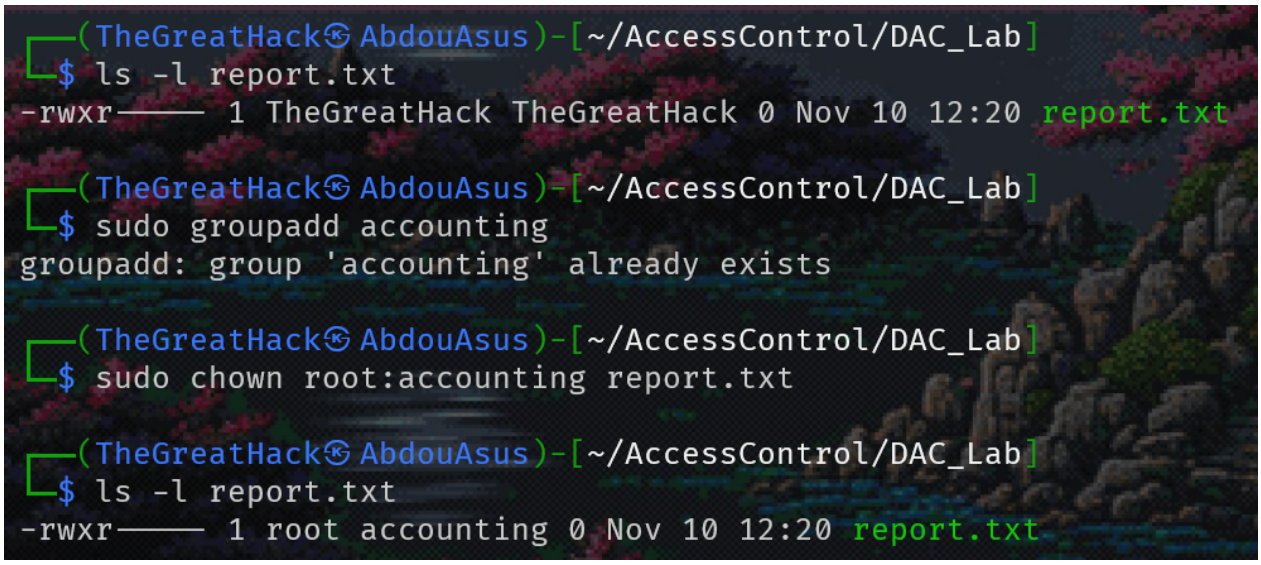


Figure 1. Creating report.txt and changing its permissions.

We changed the owner to root and the group to accounting.

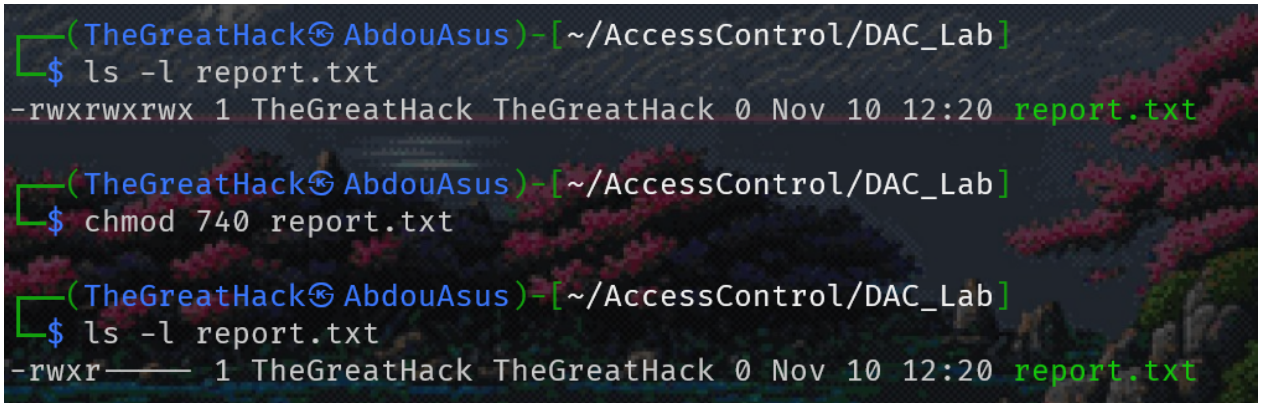


Figure 2. File ownership changed to root:accounting.

Challenge 1 - Confidential Documents

We created company/finance and example.txt inside our lab folder.

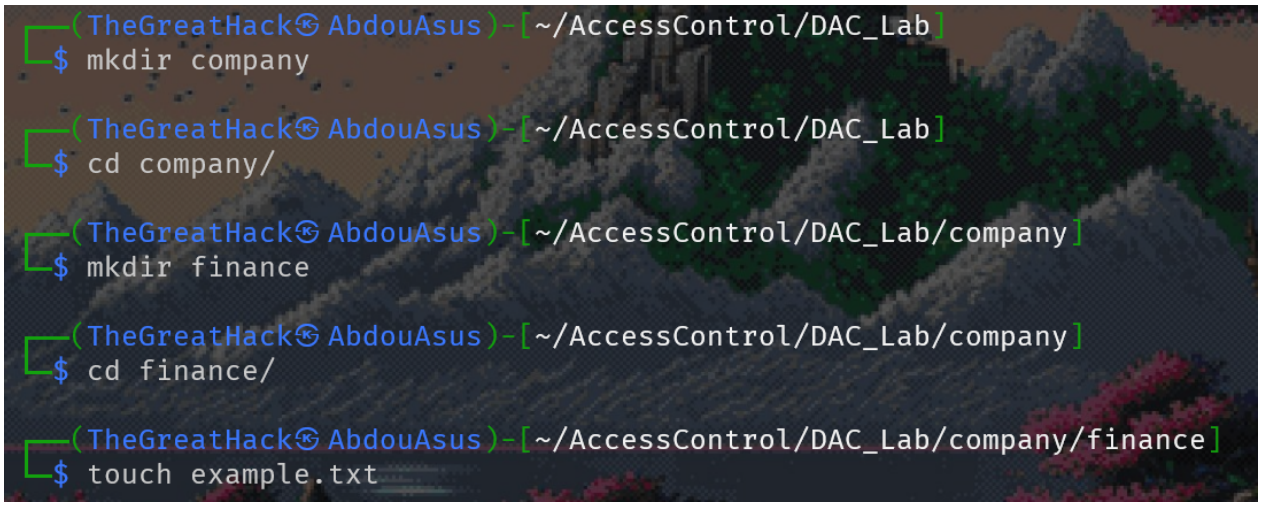


Figure 3. Creating the finance directory and example.txt.

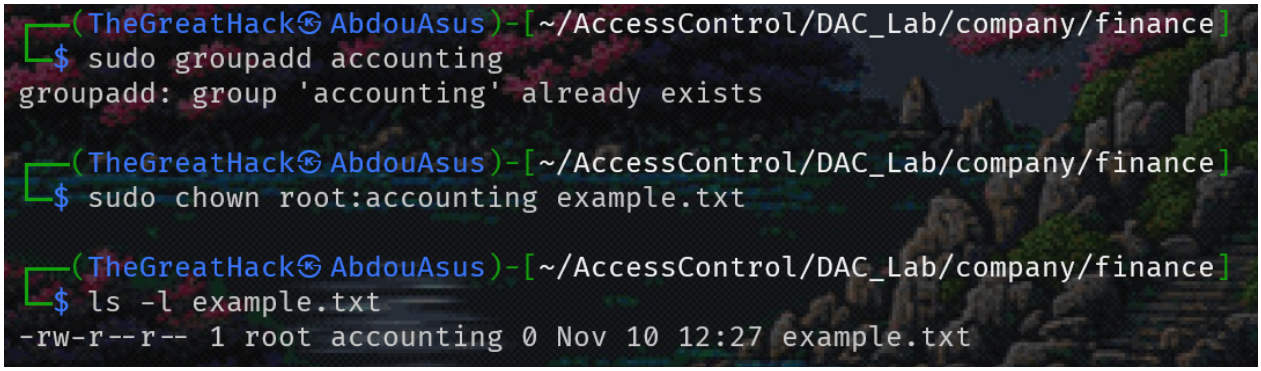


Figure 4. Setting the owner and group of example.txt.

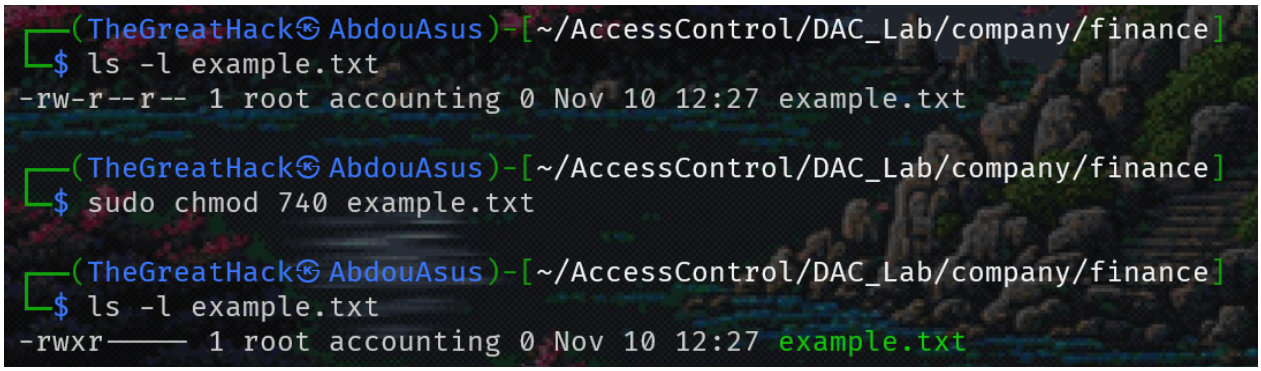


Figure 5. Setting file permissions to 740.

6. No. With directory permission 750, accounting members can enter the directory, but permission 600 on the file only allows its owner to read or write.

2. MAC - Mandatory Access Control

Bell-LaPadula uses **No Read Up** and **No Write Down**. The security levels are Secret > Confidential > Public.

Challenge 2 - Military Information System

1. Access matrix:

User	Document A (Secret)	Document B (Confidential)	Document C (Public)
Alice (Secret)	Yes	Yes	Yes
Bob (Confidential)	No	Yes	Yes
Charlie (Public)	No	No	Yes

2. No. Bob cannot read Document A because Secret is above Confidential.

3. No. Alice cannot write to Document C because it is a lower classification.

4. No Read Up prevents reading higher-level information. No Write Down prevents leaking secrets to lower levels.

5. Not necessarily. MAC does not automatically bypass ordinary Linux file permissions. Both MAC and DAC may restrict access.

3. RBAC - Role-Based Access Control

We changed the permissions of /company/source and /company/reports. The first values, 760 and 744, did not give the groups execute permission to enter the directories.



Figure 6. Initial directory permissions: source 760 and reports 744.



Figure 7. Updated directory permissions: source 770 and reports 750.

Challenge 3 - Enterprise Access Management

1. The roles are administrators, developers and viewers. They are represented by Linux groups.

2. We set source to 770 and reports to 750, with their respective groups developers and viewers.

3. The policy is not complete. Developers also need to read reports, and administrators need full access to both directories.

4. We can add these permissions using ACLs:

```
sudo setfacl -m g:developers:rx /company/reports
```

```
sudo setfacl -m g:administrators:rwx /company/source /company/reports
```

Files inside these directories must also have suitable permissions.

4. ABAC - Attribute-Based Access Control

Access is allowed when the departments match, clearance is sufficient, and the time is between 08:00 and 18:00.

Challenge 4 - Banking Access Policy

1-2. Decisions and reasons:

User	Decision	Reason
Alice	Granted	All conditions are satisfied.
Bob	Denied	HR does not match Finance.
Charlie	Denied	Public clearance is too low for Secret.
David	Denied	22:00 is outside the allowed hours.

3. Alice is denied access to an HR document because she works in Finance.

4. Yes. At 17:00 David is granted access because all conditions are satisfied.

5. ABAC is more flexible than RBAC because it considers user, document and context attributes, such as time.